

Network Level

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Content

- Network Level
 - IPv4 Problem
 - Context
 - Characteristics
 - Subnets
 - Private Networks
 - ICMP
 - Address Resolution
 - IPv6 - overview
 - Details -> Future Course

Context

- **Initial Situation**

Before the Internet, only nodes from the same network could communicate with each other

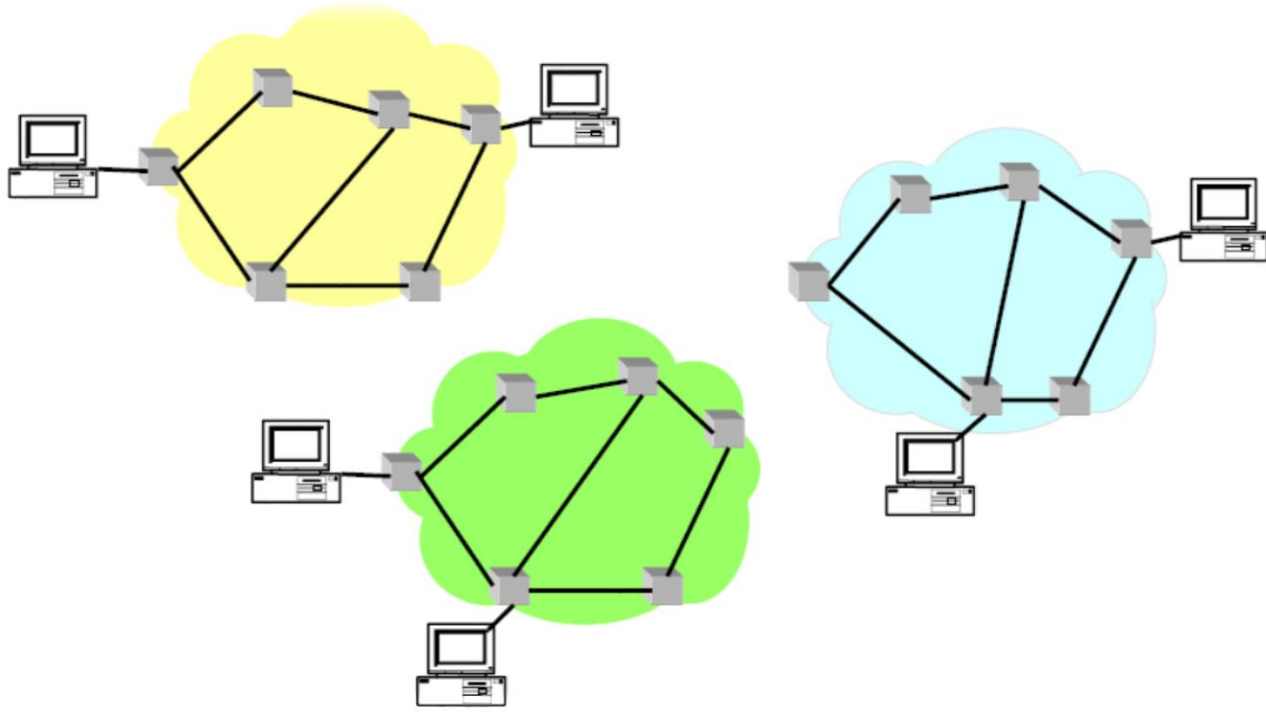
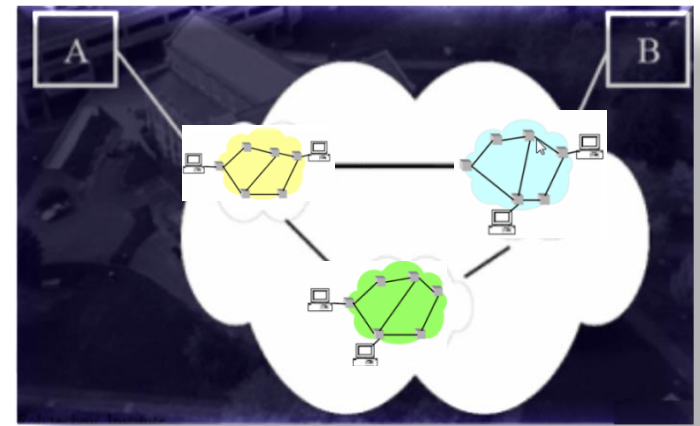


Figure: Individual Network

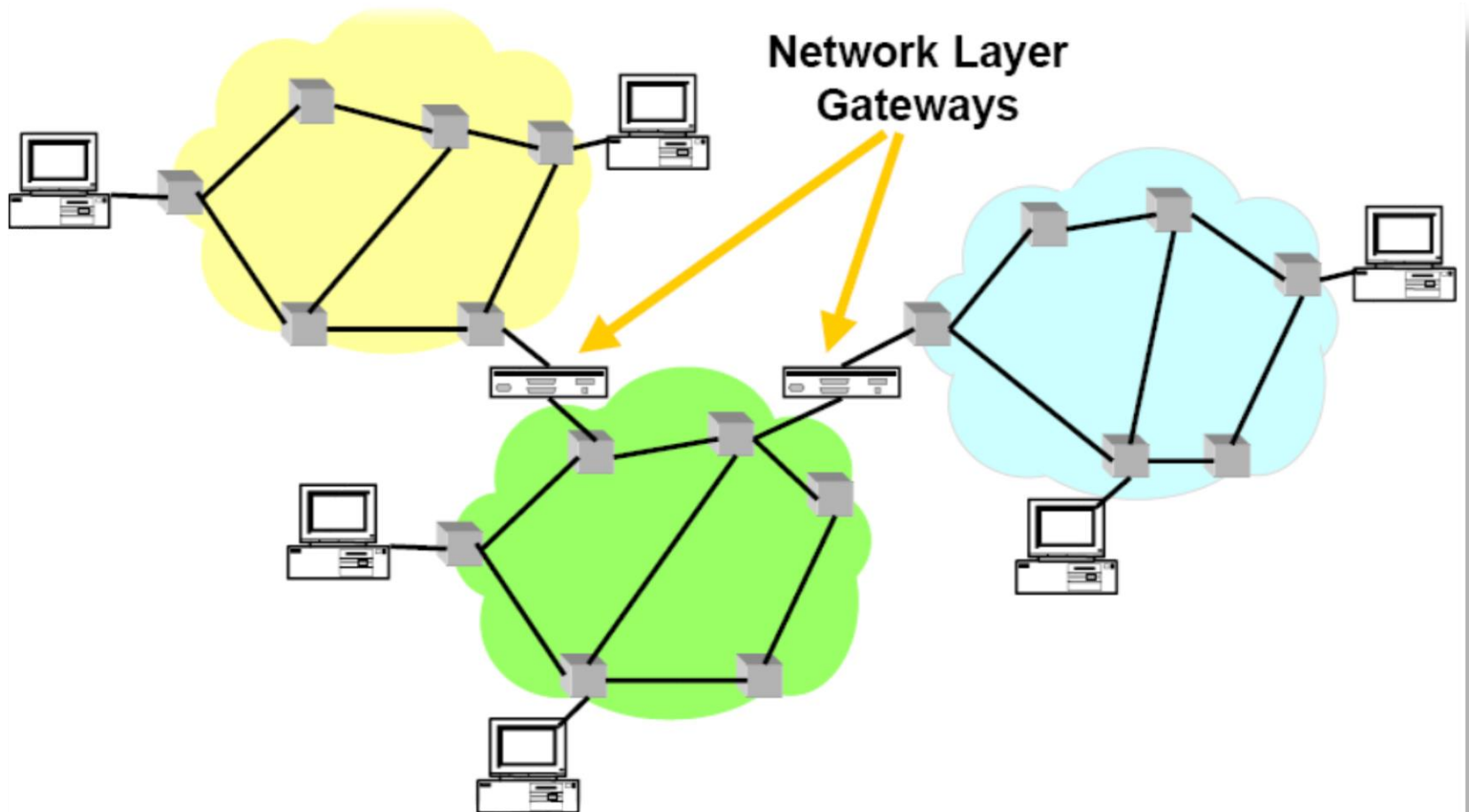
Considerations

- Problem
 - How to carry packages in a heterogeneous environment?
- **Heterogeneity**
 - At lower levels: how to make the interconnection of a large number of independent networks?
 - At higher levels: how to provide support for a wide variety of applications?
- **Scaling:** how could we handle a large number of nodes and applications in such a system of interconnected networks?



Solution

- IP – Internet Protocol



Network Level

- IP protocol is used for autonomous systems (AS - Autonomous Systems) in order to interconnect

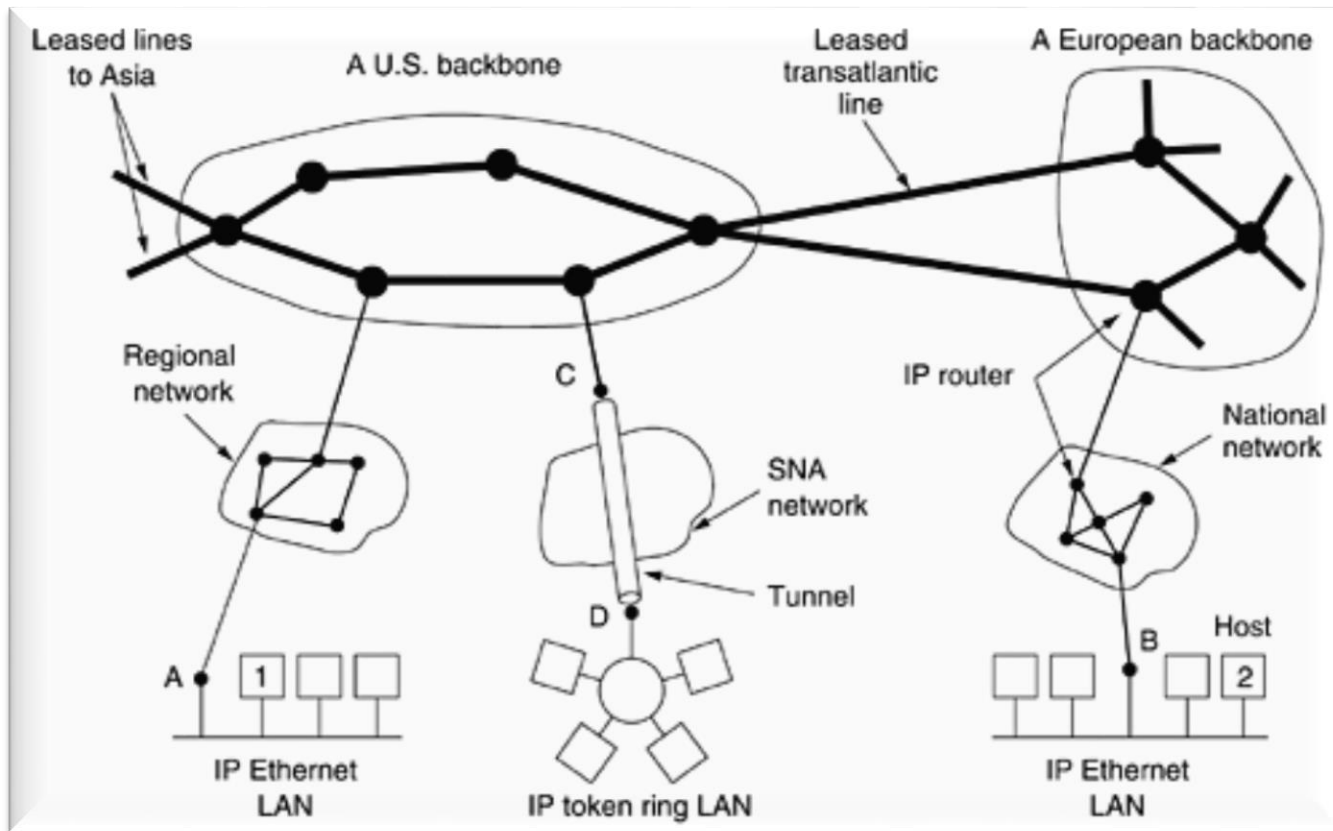


Figure: Internet - collection of interconnected networks

[Computer Networks, 2003
Andrew S. Tanenbaum]

Network Level

- Role: offers connectionless services to transport **datagrams** from source to destination; source and destination can be in different networks
- Each datagram is independent from the others
- This level does not guarantee the right transmission (loss, multiplier,...)
- +...Future Course

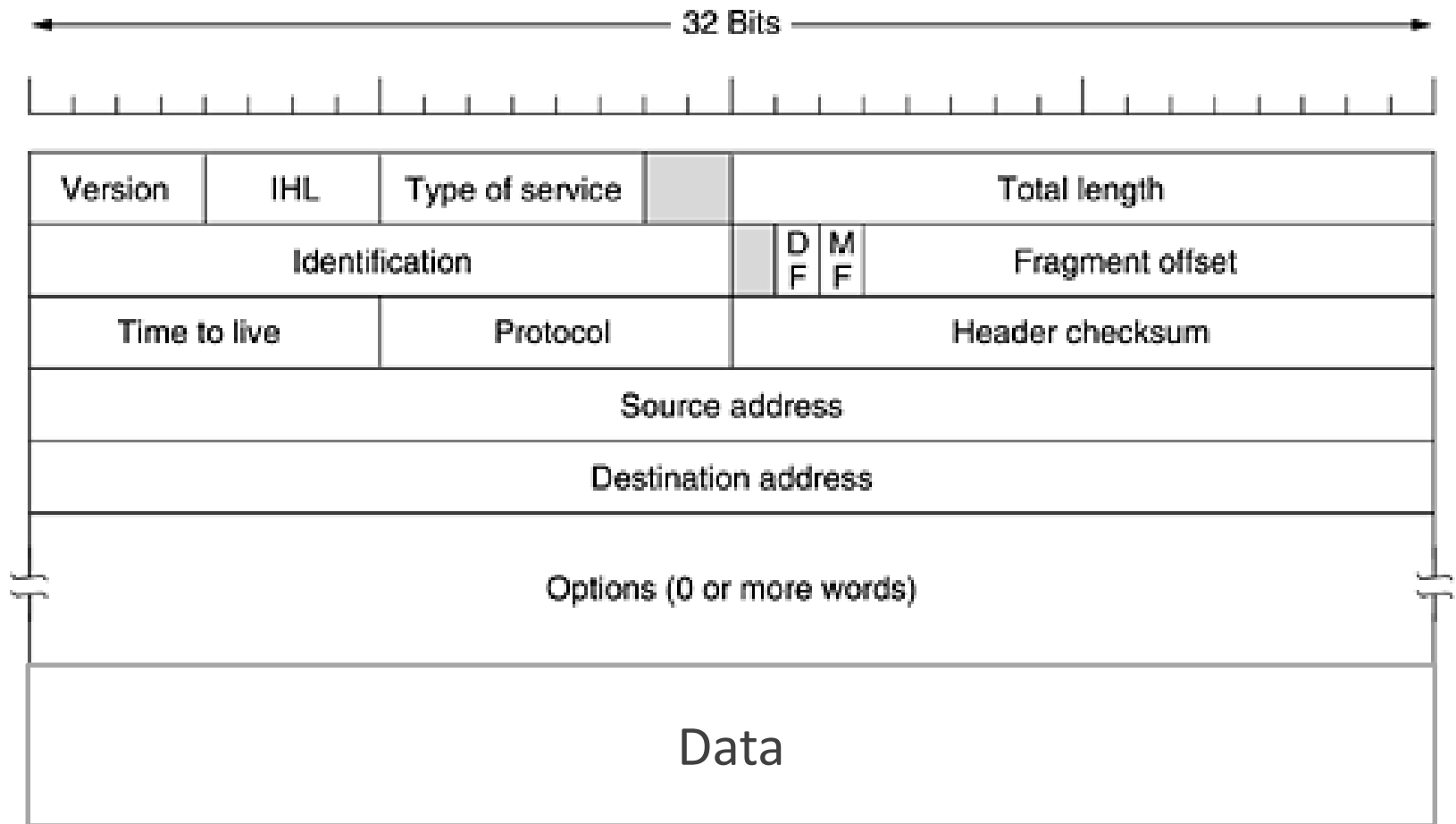
Network Level

Datagram definition

- In Internet standards **RFC 1594** a Datagram is defined as “A **self-contained**, **independent entity** of **data** carrying **sufficient information** to be **routed** from the source to the destination computer **without reliance on earlier exchanges** between this source and destination computer and the transporting network.”

IP Protocol

- IPv4 Datagram



[Computer Networks, 2003
Andrew S. Tanenbaum]

IP Protocol

- **IPv4 Datagram**

- Common values for *Version* field are:

- 4 – IP Protocol (RFC 791)
- (6 for IPv6 protocol (RFC 1883))

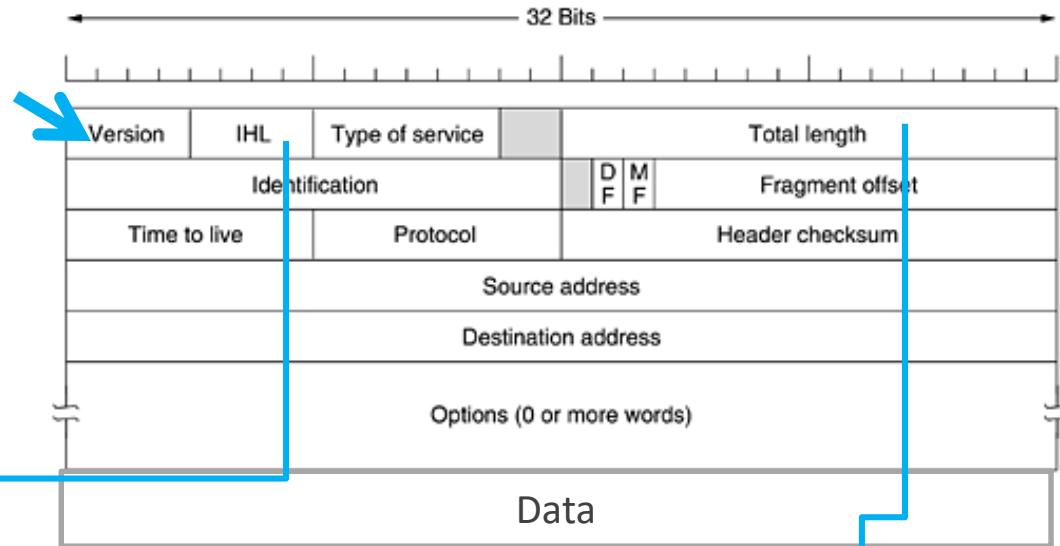


Figure: IPv4 Datagram

Specify the datagram header length

Specify the size of the entire datagram

IP Protocol

- IPv4 Datagram

- Type of service* the field allows the host to communicate to the subnet (e.g. routers) what type of service is desired

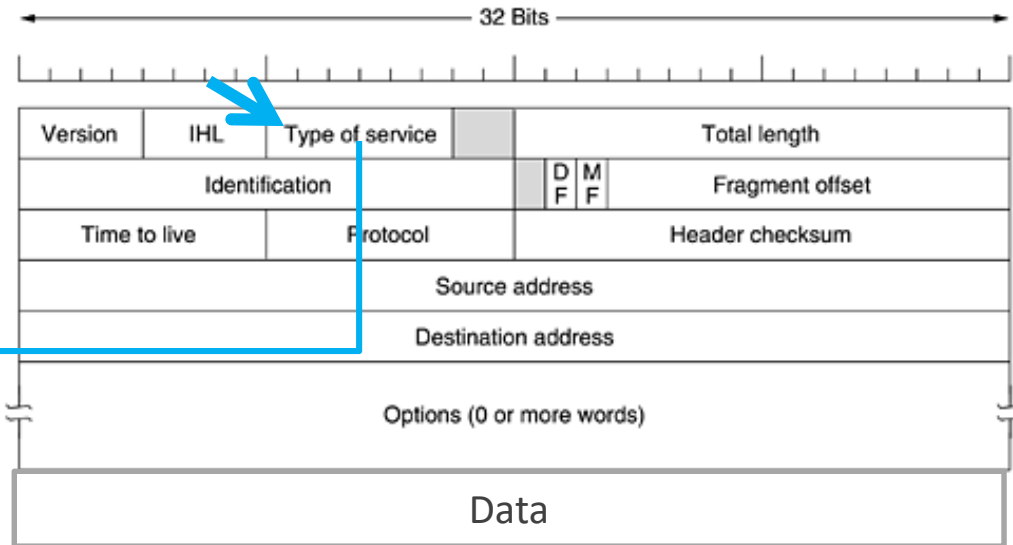


Figure: IPv4 Datagram

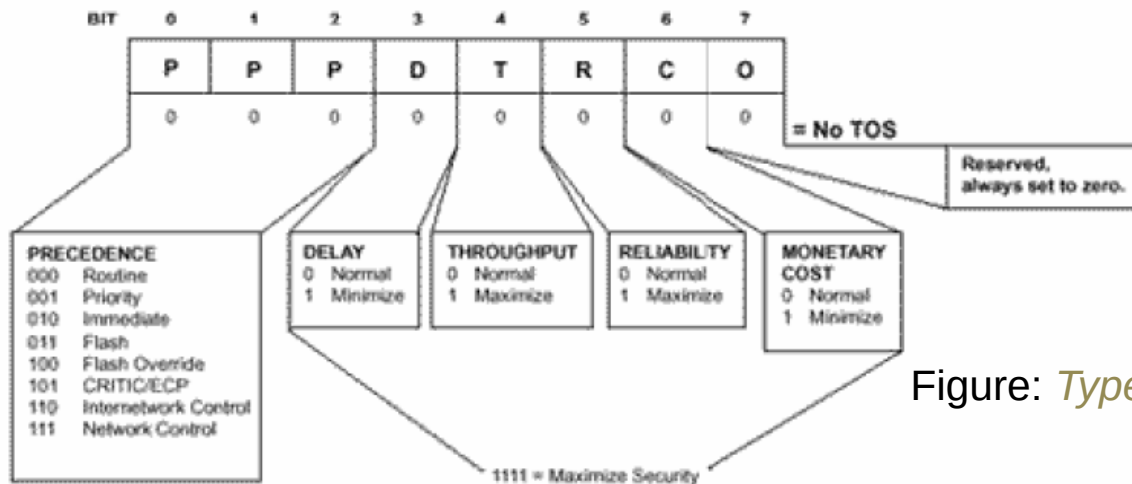


Figure: *Type of Service* Field

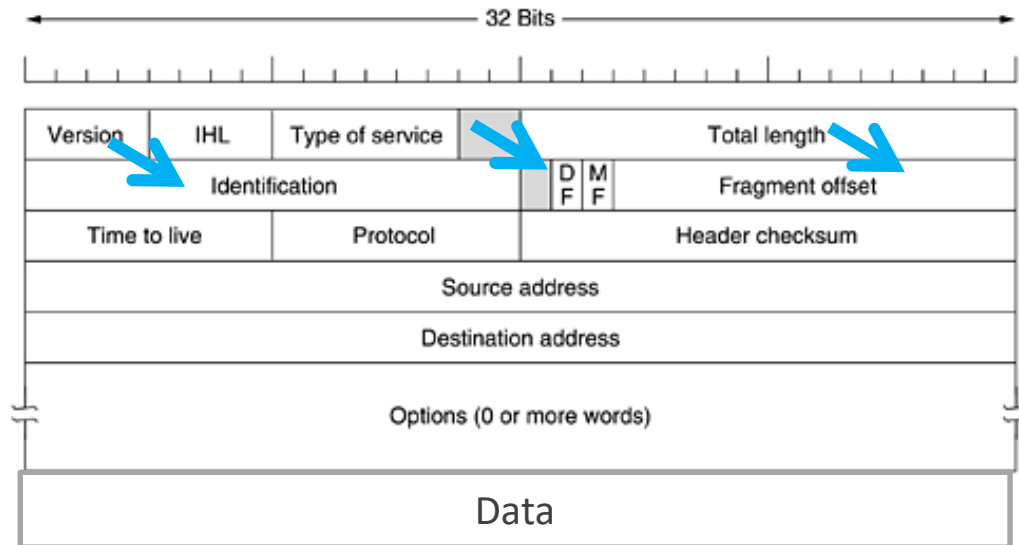
Protocolul IP

- **Datagrama IPv4**
 - *Type of service/DSCP (Differentiated Services Code Point)*
- Streaming or VoIP applications generate traffic types that have strict requirements for low latency and guaranteed bandwidth. These requirements are recognized by **QoS prioritization mechanisms** at the network and transport levels.
- **Applications** can set a priority indicator in the **IP packet header**, known as **DSCP (Differentiated Services Code Point)**, which is part of the ToS (Type of Service)/Traffic Class(IPv6) field in the IP packet header. DSCP is a mechanism used to mark packets based on their importance or priority (e.g., video, voice, or background data).

IP Protocol

- **IPv4 Datagram**

Figure: IPv4 Datagram



- Flags:
 - *DF (Don't Fragment)* – routers can't fragment the indicated datagram
 - *MF (More Fragments)* – signals that the packet is a fragment, followed by others; last fragment has this bit 0
- *Fragment offset* field - represents the fragment placed in a datagram

IP Protocol

- **IP Datagram**

- Datagram's Fragmentation:

- Each fragment has the same structure as the IP datagram
- Reassembly of datagrams is performed by the receiver
- If a fragment of a datagram is lost, the datagram is destroyed (an ICMP - Internet Control Message Protocol message is sent to the sender)
- Fragmentation mechanism has been used for some attacks - *firewall piercing* (a "special" fragment is considered as part of a connection already established, so that it can pass through a firewall)

IP Protocol

- **IP Datagram**

- Datagram Filtering:

- It is accomplished by a *firewall*: it allows access from the outside in the internal network, according to some policy, certain types of packets ((used by certain protocols / services)
 - Forestall a series of attacks regarding security
 - The firewall can be software or hardware
 - The firewall can function as a proxy or a gateway

IP Protocol

- **Proxy- role and architecture:**
 - Indirect access to other networks (Internet) to hosts on the local network (via *proxy*)
 - The proxy allows an Internet connection sharing
 - The *proxy* can be software or hardware
 - May play roles such as: *firewall* or *cache server*
 - Used to improve the performance (e.g., caching, flow control), filtering messages, ensuring anonymity

IP Protocol

- **IPv4 Datagram**

- *TTL (Time to Live)* field specifies the lifetime of the package;
the number is decremented by every router through which the packet passes

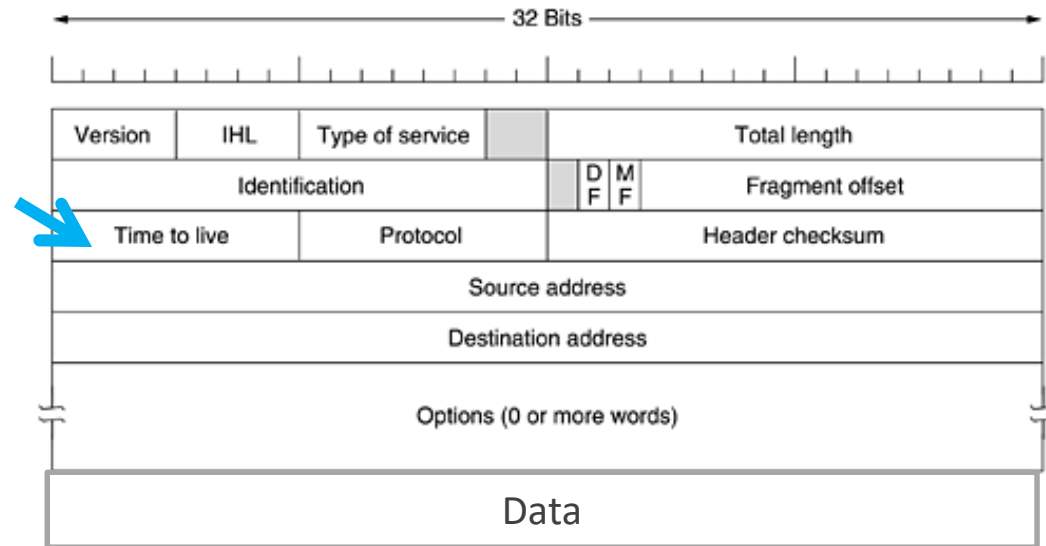


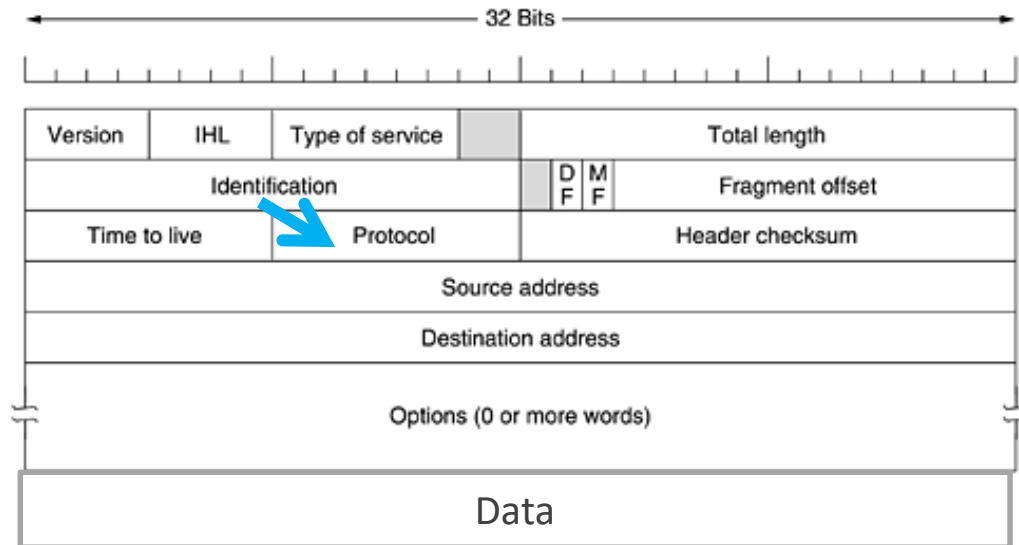
Figure: IPv4 Datagram

IP Protocol

- **IPv4 Datagram**

- *Protocol* field specifies the protocol (from the superior level) intended to process the datagram

Figure: IPv4 Datagram



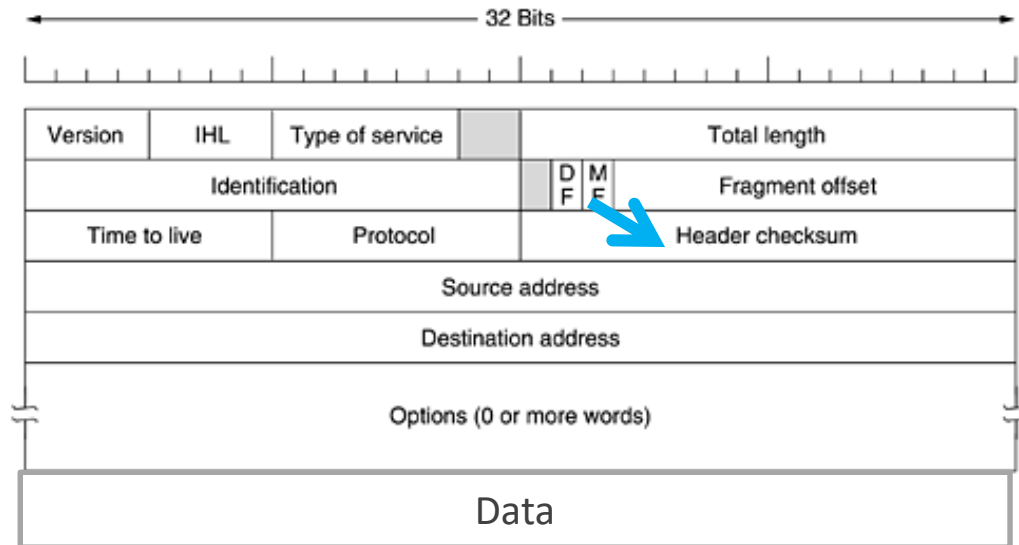
- 1 ICMP (*Internet Control Message Protocol*)
- 2 IGMP (*Internet Group Message Protocol*)
- 6 TCP (*Transmission Control Protocol*)
- 17 UDP (*User Datagram Protocol*)
- ... etc.(RFC 1700)

IP Protocol

- **IPv4 Datagram**

- *Header checksum field* used for detection; if an error occurs the datagram is destroyed

Figure: IPv4 Datagram



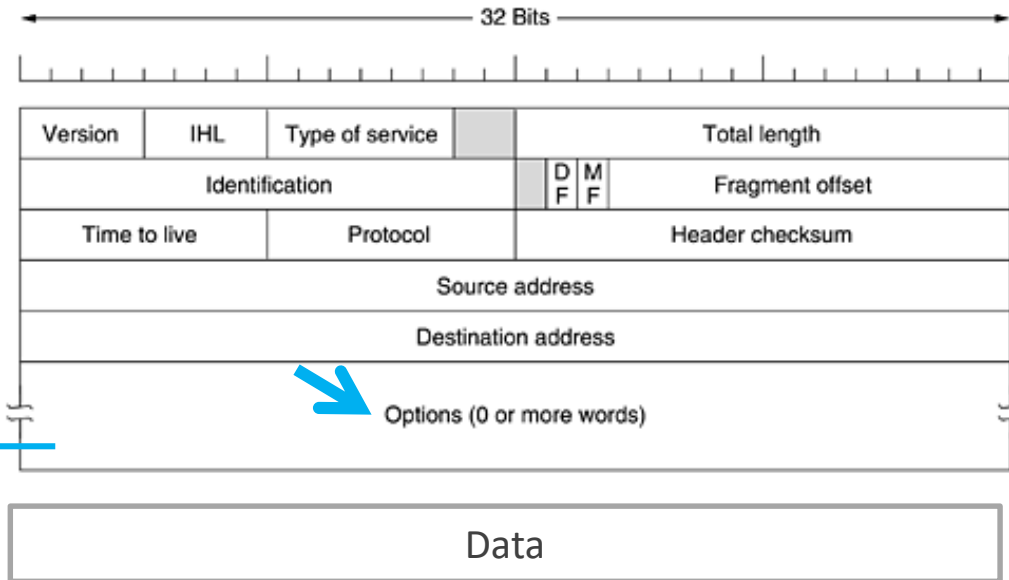
Header Checksum (16 bits)

- **Errors** are checked by the Header Checksum at **each router**
 - Checking is needed because IP header fields **may change** when the packet is sent out a router (e.g., TTL, Flags, Fragmentation, etc.)

IP Protocol

Figure: IPv4 Datagram

- **IPv4 Datagram**
 - *Options* Field



Options	Details
Security	Mention if the datagram is a “secret” one
Strict source routing	Show full path to go
Loose source routing	Indicates a list of routers that should not be skipped
Record route	Each router adds its own IP
Timestamp	Each router adds its own IP and a timestamp

IP Protocol

- **IPv4 Datagram**

- *Source address* and *Destination address* fields indicate the source address and the destination address

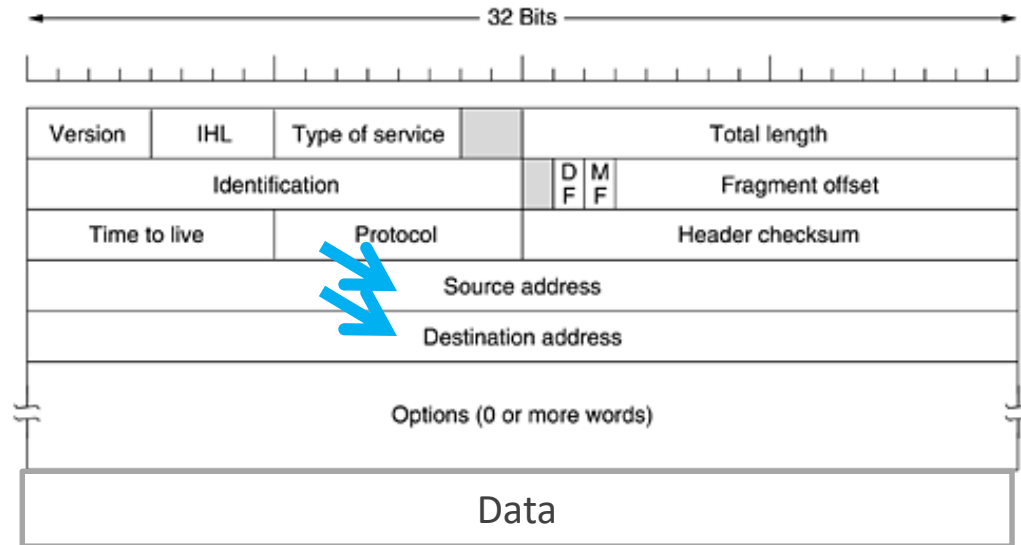


Figure: IPv4 Datagram

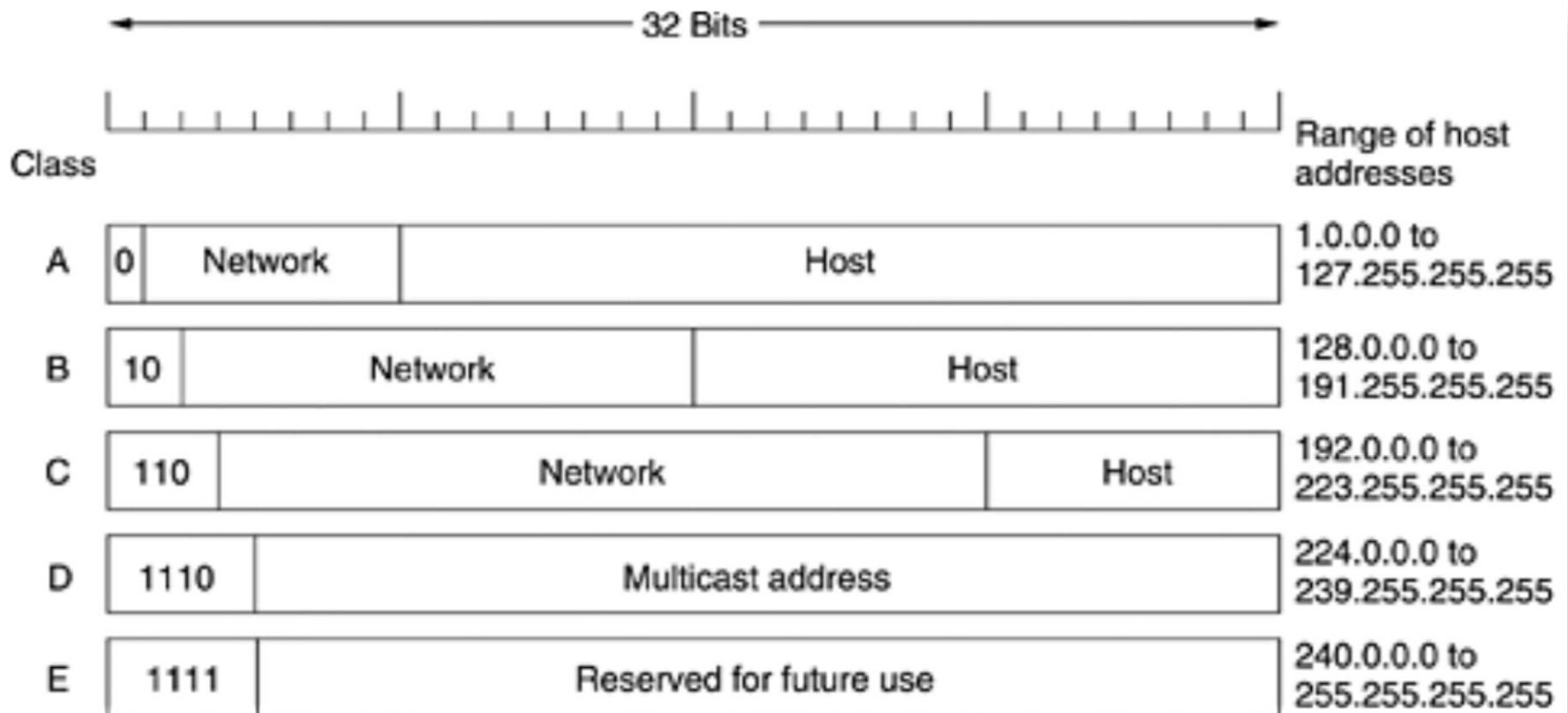
IP Protocol

- **IPv4 Addresses**

- Each IP address includes a network identifier(**NetID**) and a **host identifier (HostID)**
- Each network interface has a single IPv4 address
- An IPv4 address has a length of 32 bits
- Initially (RFC 791) there was a division into network classes: A,B,C,D,E

IP Protocol

- IPv4 Addresses



[Computer Networks, 2003
Andrew S. Tanenbaum]

IP Protocol

- **IPv4 Addresses**

- **Class A:** 128 possible networks, 2^{24} hosts/network
- **Class B:** 2^{14} possible networks, 2^{16} hosts/network
- **Class C:** over 2 million networks, 255 hosts/network
- **Network Identifier(NetID)** is assigned by a central authority (NIC – *Network Information Center*)
- **Host Identifier(HostID)** is assigned locally by a network administrator
- Example: 85.122.23.145 – Class A (in decimal notation convention)
0101 0101 0111 1010 0001 0111 1001 0001
- For IPv6, hexadecimal representation is recommended

IP Protocol

- **IPv4 Addresses**

- An interface network has assigned a unique IP address
- A host can have multiple NICs, therefore it has multiple IP addresses
- The hosts of the same network have the same network identifier (the same NetID)
- *Broadcast* addresses have HostID's bites equaled to 1
- The IP address in which all HostID's bites are 0 is called a *network address* – refers to the hole network
 - Example: 85.122.23.0 (*network address* for a host such us 85.122.23.145 and 85.122.23.1)
- 127.0.0.1 – *loopback address (localhost)*

IP Protocol

- **IPv4 Addresses**

- From the address space, some addresses are reserved: (RFC 1918):
 - 0.0.0.0 – 0.255.255.255
 - 10.0.0.0 – 10.255.255.255 (private addresses)
 - 127.0.0.0 – 127.255.255.255 (*loopback* addresses)
 - 172.16.0.0 - 172.31.255.255 (private addresses)
 - 192.168.0.0 - 192.168.255.255 (private addresses)
- Private addresses : addresses that are not accessible to the outside (the "real" Internet), but only in the organization's intranet

Private Networks

- Aspects:
 - The exponential growth of the host number
 - Not all hosts offer resources available on the Internet
 - Many waste IP Addresses
- CIDR(Classless Inter-Domain Routing)
 - Method to efficiently allocate IP Addresses and routing
 - Replace former classification (e.g. A, B,C)
 - Makes the Internet more scalable $\Leftrightarrow$ networks can be assigned proper subnet sizes
 - Uses *VLSM (Variable-Length Subnet Masking)*
 - *E.g. 123.234.100.56/24* sau *2001:db8::/32 (CIDR notation)*

iana.org/numbers

Number Resources

We are responsible for global coordination of the Internet Protocol addressing systems, as well as the Autonomous System Numbers used for routing Internet traffic.

Currently there are two types of Internet Protocol (IP) addresses in active use: IP version 4 (IPv4) and IP version 6 (IPv6). IPv4 was initially deployed on 1 January 1983 and is still the most commonly used version. IPv4 addresses are 32-bit numbers often expressed as 4 octets in “dotted decimal” notation (for example, 192.0.2.53). Deployment of the IPv6 protocol began in 1999. IPv6 addresses are 128-bit numbers and are conventionally expressed using hexadecimal strings (for example, 2001:0db8:582:ae33::29).

Both IPv4 and IPv6 addresses are generally assigned in a hierarchical manner. Users are assigned IP addresses by Internet service providers (ISPs). ISPs obtain allocations of IP addresses from a local Internet registry (LIR) or National Internet Registry (NIR), or from their appropriate Regional Internet Registry (RIR):



REGISTRY	AREA COVERED
AFRINIC	Africa Region
APNIC	Asia/Pacific Region
ARIN	Canada, USA, and some Caribbean Islands
LACNIC	Latin America and some Caribbean Islands
RIPE NCC	Europe, the Middle East, and Central Asia

Our primary role for IP addresses is to allocate pools of unallocated addresses to the RIRs according to their needs as described by [global policy](#) and to document protocol assignments made by the [IETF](#). When an RIR requires more IP addresses for allocation or assignment within its region, we make an additional allocation to the RIR. We do not make allocations directly to ISPs or end users except in specific circumstances, such as allocations of multicast addresses or other protocol specific needs.

- Internet Assigned Numbers Authority

<https://www.iana.org>

IP Protocol

- **Subnets using network masks**

- It appeared as a solution to the problem of IP address space's exhaustion
- Simplify Routing
- Subnets cannot be detected externally

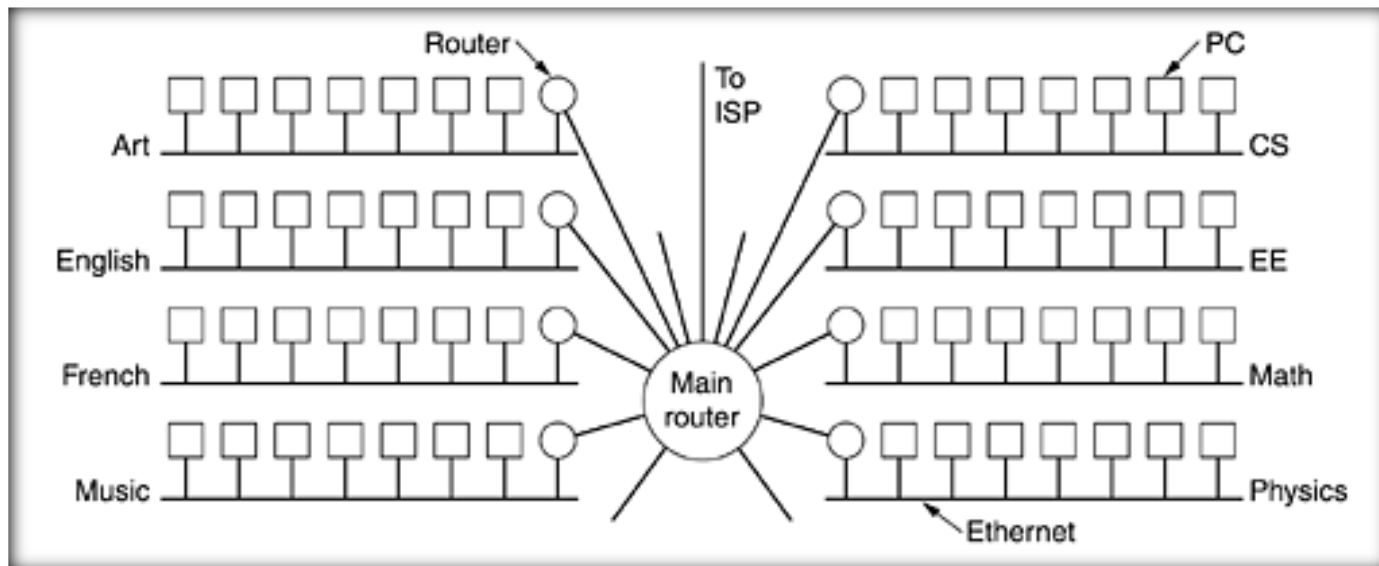
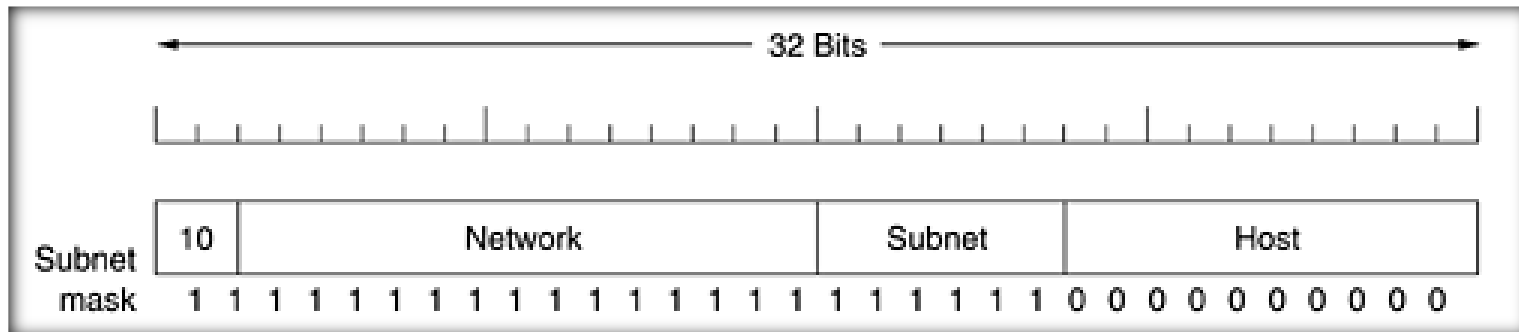


Figure: A campus network

IP Protocol

- **Subnets using network masks**

- Division into subnetworks will be made via the network mask(*netmask*): NetID bits are 1, HostID bits are 0
- Subnet identifier (**SubnetID**) is generally used to group computers based on physical topology



Example. One way to create a subnet in a B network

IP Protocol

- **Subnets using network masks**

- Example:

- Let's consider the IP address: 160.0.6.7

- 10100000 00000000 00000110 00000111

- Network mask: 255.255.252.0

- 11111111 11111111 11111100 00000000



- Network address: 160.0.4.0

- 10100000 00000000 00000100 00000000

Network address = network mask AND IP address

- Default subnet masks:

- 255.0.0.0 - Class A

- 255.255.0.0 - Class B

- 255.255.255.0 - Class C

IP Protocol

123.234.100.56/24 represents

- IPv4 address (32 bits) 123.234.100.56
- Subnet mask with 24 ones
= 11111111.11111111.11111111.00000000
= 255 . 255 . 255 . 0
- Subnet size = $2^{(32-24)} = 2^8 = 256$ IPv4 addrs.
in this subnet
- Routing prefix 123.234.100.0 can be
obtained from IPv4 address 123.234.100.56
and Subnet mask 255.255.255.0

IP Protocol

- **Convention mark:** x.x.x.x/m means that we apply an m bits mask to the IP specified x.x.x.x address
- Example:
 - 10.0.0.0/12 – it applies a 12-bits mask to 10.0.0.0 address, we select possible values for the last 20 bits ($=32-12$)
 - 85.122.16.0/20 – it applies a 20-bit mask to 85.122.16.0 address

Private Networks

- Aspects:
 - The exponential growth of the hosts number
 - Not all hosts offer resources available on the Internet
- Solution: NAT (*Network Address Translation*) – RFC 3022, 4008
- The private addresses can be reused (RFC 1918)
 - It is based on replacing the private IP address with a public IP address (*IP masquerading*)

Private Networks

- Functionality:

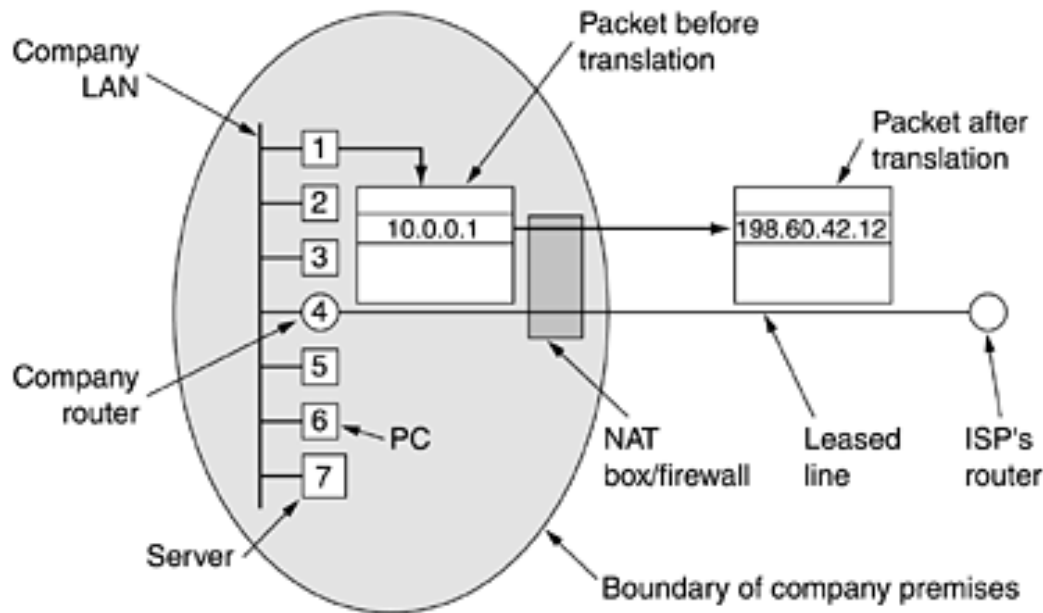


Figure: NAT mechanism

- Routers normally ignore datagrams containing private addresses => private IP addresses can be used in the organization's intranet
- Access to the outside (the "real" Internet) is achieved via a gate (**mediating gateway**) that rewrites the source IP addresses / destination

Network Level

- Protocols
 - ICMP (RFC 792)
 - ARP (RFC 826)
 - RARP (RFC 903)
 - BOOTP (RFC 951,1048,1084)
 - DHCP
- From IPv4 to IPv6

ICMP Protocol

- **ICMP – Internet Control Message Protocol**

- Used to exchange control messages
- Use IP
- ICMP messages are processed by the IP software, not by the user processes
- Messages types

Message Type	Description
8 Echo Request	Ask if a host is active
0 Echo Replay	“Yes, I’m active”
3 Destination Unreachable	The package can’t be delivered (e.g. DF is set)
5 Redirect	The message is not correctly routed
11 Time Exceeded	Time elapsed (TTL=0) <- (e.g. loop, congestions, low values for time)
... etc (RFC 792)	http://www.iana.org/assignments/icmp-parameters

Protocolul ICMP

- Used by:
 - **ping** command (Packet Internet Gropher)
 - **traceroute** command
 - A package with TTL=1 (1 hop) is sent
 - The first router ignores the packet and sends back an ICMP message *"time-to-live exceeded"*
 - A package with TTL=2 is sent (2 hops)
 - The second router ignores the packet and sends back an ICMP message *"time-to-live exceeded"*
 - Repeat until it has received a response from the destination or has reached the maximum number of hops

Address resolution

- **IP addresses <-> hardware addresses (physical)**
 - The process of finding the hardware address of a host, knowing its IP address is called address resolution(*address resolution*) – **ARP** protocol (RFC 826) & Neighbor Discovery Protocol (NDP)-IPv6
 - ARP –broadcast protocol (each host receives a request for a physical address, and the answer is given by the one in question)
 - The process of finding the IP address based on the hardware address is called *reverse address resolution* –**RARP** Protocol (RFC 903)
 - Used to boot workstations without disks
 - **BOOTP** (RFC 951,1048,1084)
 - **DHCP** (*Dynamic Host Configuration Protocol*) RFC 2131,2132

IPv6

- Context:
 - Issues in IPv4 addresses world:
 - The exponential growth of the hosts` number
 - Very large routing tables
 - Complex configurations, more and more users (and increasing)
 - Hard to ensure QoS
 - Pressure from mobile operators

IPv6

- Objectives for a new protocol:
 - Support for billions of hosts
 - Reducing routing tables
 - Simplifying Protocol
 - Support for mobile hosts
 - Compatibility with the old IP
 - Support for future developments of the Internet
 - RFC 2460, 2553

IPv6



- 6 June 2012

IPv6

- **Aspects:**

- IPv6 addresses are 16 bytes in length - 2^{128} addresses
- Note: 8 groups of 16 bits (=4 hexadecimal digits) separated by “:”
 - Example: 2001:0db8:0000:0000:0000:0000:1428:57ab
 - Zeros in front of the 4 hexadecimal digits can be omitted
 - If one or more groups of 4 digits is 0000, the zeros may be omitted and replaced (once) with “::”
 - Example: 2001:db8::1428:57ab
- To maintain compatibility, public IP addresses are considered a subset of IPv6 address space
- IPv4 addresses in IPv6 can be written as: 10.0.0.1 -> ::10.0.0.1 or 0:0:0:0:0:0:A00:1

IPv6

- IPv6 Address Length: **128** bits
 - **8** groups of 16 bits (= 4 hexadecimal digits)
 - 8 groups x 4 hexadecimal digits x 4 bits/hexadecimal
 - 2ac1:05b8:123e:0000:0000:0000:03a0:c234
 - **1 : 2 : 3 : 4 : 5 : 6 : 7 : 8** groups

Binary = **Hexa** = **Decimal**

- 0000 = 0x0 = 0
- 0001 = 0x1 = 1
- 0010 = 0x2 = 2
- 0011 = 0x3 = 3
- 0100 = 0x4 = 4
- 0101 = 0x5 = 5
- 0110 = 0x6 = 6
- 0111 = 0x7 = 7

Binary = **Hexa** = **Decimal**

- 1000 = 0x8 = 8
- 1001 = 0x9 = 9
- 1010 = 0xa = 10
- 1011 = 0xb = 11
- 1100 = 0xc = 12
- 1101 = 0xd = 13
- 1110 = 0xe = 14
- 1111 = 0xf = 15

IPv6

- ICMPv6
 - ICMP provides functions (reporting data transmission, errors, etc.) plus:
 - Neighbor Discovery(*Neighbor Discovery Protocol – NDP*) - Replaces the ARP
 - Multicast listener discovery(*Multicast Listener Discovery*) – replaces IGMP (*Internet Group Management Protocol*)
 - Details in RFC 4443

IPv6

- ... More -> Optional Course

Summary

- Network Level
 - IPv4 Problem
 - Context
 - Characteristics
 - Subnets
 - Private Networks
 - ICMP
 - Address Resolution
 - IPv6 - overview
 - Details -> Future Course



Questions?